Electromagnetic Induction

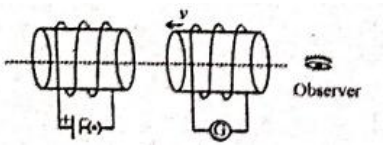
1. According to Faraday's Law of electromagnetic induction an emf is induced in a coil if
1) electric flux links with the coil
2) magnetic flux links with the coil
3) magnetic flux links with the coil changes
4) None of the above

2. A coil and a magnet moves with their constant speed 5 m/sec. and 3 m/sec respectively, towards each other. Induced emf in coil is 16 mV. If both moves in same direction, then induced emf in coil:
1) 15 mV
2) 4 mV
3) 64 mV
4) Zero

3. The radius of a circular coil having 50 turns in 2 cm. Its plane is normal to the magnetic field. The magnetic field changes from 2T to 4T in 3.14 sec. The induced emf in coil will be:
1) 0.4V
2) 0.04V
3) 4 mV
4) 0.12 v

4. Magnetic field changes at the rate of 0.4 T/sec in a square coil of side 4 cm kept perpendicular to the field. If the resistance of the coil is $2\times10^{-3}\Omega$,. then induced current in coil is
1) 0.16 A
2) 0.32 A
3) 3.2 A
4) 1.6 A

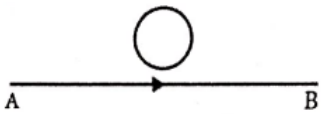
5. The current flows in a circuit as shown below. If a second circuit is brought near the first circuit then the current in the second circuit will be:



1) Clock wise
2) Anti clock wise
3) Depending on the value of R_G
4) None of the above

6. A coil of resistance 10Ω and 1 turns have the magnetic flux line of $5.5 \times 10^{-4} Wb$. If the magnetic flux changed to $5 \times 10^{-4} Wb$ in 0.1 sec, then the induced charge in coil is:
1) 50 μC
2) 5 μC
3) 2 μC
4) 20 μC

7. The current flows from A to B as shown in the figure. The direction of the induced current in the loop is:



1) Clockwise
2) Anticlockwise
3) Straight line
4) None of these

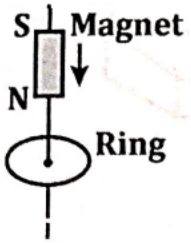
8. If number of turns of 70 cm² coil is 200 and it is placed in a magnetic field of 0.8 Wb/m² which is perpendicular to the plane of coil and it is rotated through an angle 180° in 0.1 sec, then induced emf in coil

1) $11.2\,V$
2) $1.12\,V$
3) $22.4\,V$
4) $2.24\,V$

9. A coil of mean area 500 cm² and having 1000 turns is held perpendicular to a uniform field of 0.4 gauss. The coil is turned through 180° in $\frac{1}{10}$ second. The average induced emf.

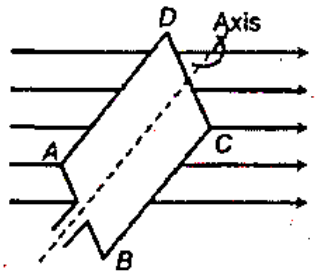
1) $0.04\,V$
2) $0.4\,V$
3) $4\,V$
4) $0.004\,V$

10. Consider a metal ring kept on a horizontal plane. A bar magnet is held above the ring with its length along the axis of the ring. If the magnet is dropped freely the acceleration of the falling magnet is (g is acceleration due to gravity)



1) More than g
2) Equal to g
3) Less than g
4) Depend on mass of magnet

11. A rectangular coil ABCD is rotated anticlockwise with a uniform angular velocity about the axis shown in diagram below. The axis of rotation of the coil as well as the magnetic field B are horizontal. The induced emf in the coil would be minimum when.



1) The plane of the coil is horizontal
2) The plane of the coil makes an angle of 45° with the magnetic field
3) The plane of the coil is at right angles to the magnetic field
4) The plane of the coil makes an angle of 30° with magnetic field

12. A 10 metre kept in east-west falling with velocity 5m/sec perpendicular to the field $0.3\times10^{-4}\,Wb/m^2$. The induced e.m.f. across the terminal will be.

1) $0.15\,V$
2) $1.5\,mV$
3) $1.5\,V$
4) $15.0\,V$

13. When a wire loop is rotated in a magnetic field, the direction of induced e.m.f. change once in each.

1) $\frac{1}{4}$ revolution
2) $\frac{1}{2}$ revolution
3) 1 revolution
4) 2 revolution

14. A metal conductor of length 1 m rotates vertically about one of its ends at angular velocity 5 radians per second. If the horizontal component of earth's magnetic field is $0.2\times10^{-4}\,T$, then the e.m.f. developed between the two ends of the conductor is

1) $5\,mV$
2) $5\times10^{-4}\,V$
3) $50\,mV$
4) $50\,\mu V$

15. A conducting rod of length I is falling with a velocity v perpendicular to uniform horizontal magnetic field B. The potential difference between its two ends will be.

1) $2\,Blv$
2) Blv
3) $\frac{1}{2}Blv$
4) $B^2l^2v^2$

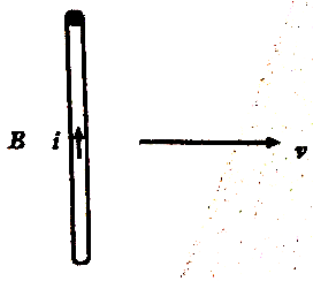
16. A coil of N turns and mean crosssectional area A is rotating with uniform angular velocity ω about an axis at right angle to uniform magnetic field B. The induced e.m.f. E in the coil will be.

1) $NBA\sin\omega t$
2) $NB\omega\sin\omega t$
3) $NB/A\sin\omega t$
4) $NBA\omega\sin\omega t$

17. A copper disc of radius 0.1 m is rotated about its centre with 20 revolutions per second in a uniform magnetic field of 0.1 Tesla with its plane perpendicular to the field. The e.m.f. induced across the radius of disc is.

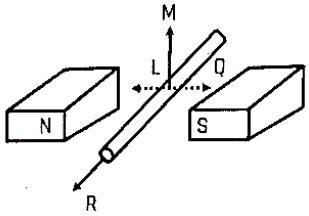
1) $\frac{\pi}{10}V$
2) $\frac{2\pi}{10}V$
3) $\pi\times10^{-2}\,V$
4) $20\pi\times10^{-3}\,V$

18. A conducting wire is moving towards right in a magnetic field B. The direction of induced current in the wire is shown in the figure. The direction of magnetic field will be.



1) In the plane of paper pointing towards right
2) In the plane of paper pointing towards left
3) Perpendicular to the plane of paper and down-wards
4) Perpendicular to the plane of paper and upwards

19. An electric potential difference will be induced between the ends of the conductor shown in the diagram, when the conductor moves in the direction.

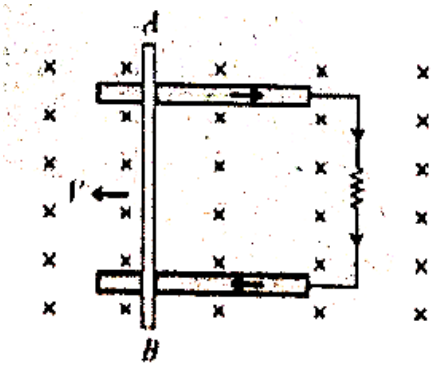


1) R
2) Q
3) L
4) M

20. A straight conductor of length 0.4 m is moved with a speed of 7 m/s perpendicular to the magnetic field of intensity of 0.9 Wb/m. The induced e.m.f. across the conductor will be.

1) 7.25 V
2) 2.75 V
3) 1.25 V
4) 2.52 V

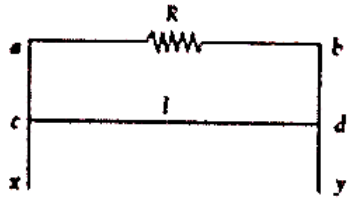
21. Consider the situation shown in the figure. The wire AB is sliding on the fixed rails with a constant velocity. If the wire AB is replaced by semicircular wire, the magnitude of the induced current will:



1) Increase
2) Remain the same.
3) Decrease
4) None of the above

22. A wire cd of length l and mass m is sliding without friction on conducting rails ax and by as shown. The vertical rails are connected to each other with a resistance R between a and b. A

uniform magnetic field B is applied perpendicular to the plane $abdc$ such that cd moves with a constant velocity of:



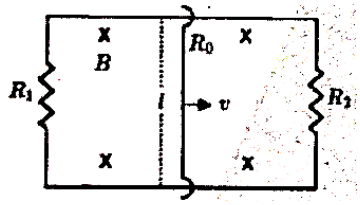
1) $\frac{mgR}{Bl}$

2) $\frac{mgR}{B^2l^2}$

3) $\frac{mgR}{B^3l^3}$

4) $\frac{mgR}{B^2l}$

23. A rod having length l and resistance R_0 is moving with speed v as shown in the figure. The current through the rod is:



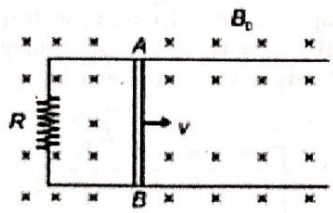
1) $\frac{Blv}{\frac{R_1R_2}{R_1+R_2}+R_0}$

2) $\frac{Blv}{\left(\frac{1}{R_1}+\frac{1}{R_2}+\frac{1}{R_0}\right)^2}$

3) $\frac{Blv}{R_1+R_2+R_0}$

4) $\frac{Blv}{\frac{1}{R_1}+\frac{1}{R_2}+\frac{1}{R_0}}$

24. A rod AB of length is moving with constant speed v in a uniform magnetic field on a conducting U-shaped wire as shown. If the rate of loss of heat energy across resistance R is Q. then the force needed parallel to velocity to keep rod moving with constant speed v is;



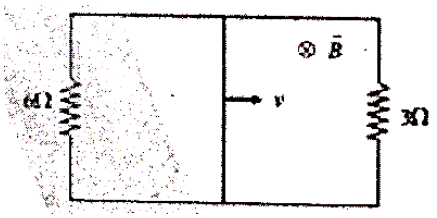
1) Qv

2) $\frac{Q}{v}$

3) $\frac{Q^3}{v}$

4) $Q^2 v$

25. A rectangular loop with a sliding connector of length $l = 1.0\,m$ is situated in a uniform magnetic field $B = 2T$ perpendicular to the plane of the loop. Resistance of connector is $r = 2\Omega$. Two resistances of 6Ω and 3Ω are connected as shown in the figure. The external force required to keep the connector moving with a constant velocity $v = 2m/s$ is:

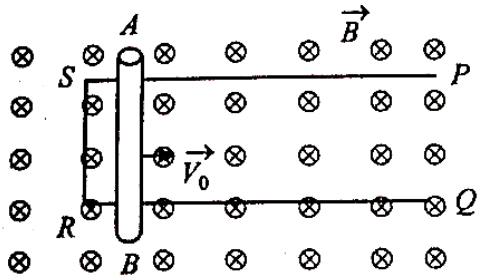


1) 6 N
2) 4 N
3) 2 N
4) 1 N

26. A conductor of length 100 cm, moves at right angle to a uniform field flux density of $1.5\,wb/m^2$ with a velocity of 50m/s. The emf induced in the conductor will be:

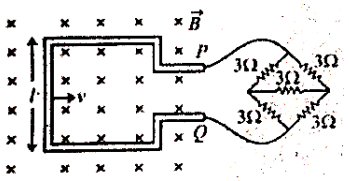
1) 150 V
2) 75 V
3) 50 V
4) 37.5 V

27. A conducting rod AB of length l s projected on a frictionless frame $PSRQ$ with velocity V_0 at any instant. The velocity of the rod after time t is



1) $V = V_0$
2) $V > V_0$
3) $V < V_0$
4) None of these

28. A square metallic wire loop of side 0.1 m and resistance of 1Ω is moved with a constant velocity in a magnetic field of 2wb/m² as shown in the figure. The magnetic field is perpendicular to the plane of the loop and the loop is connected to a network of resistances. What should be the velocity of the loop so as to have a steady current 1 mA in the loop?



1) $1\,cm/\sec$

2) $2\,cm/\sec$

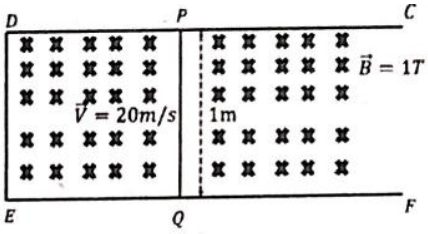
3) $3\,cm/\sec$

4) $4\,cm/\sec$

29. An aero plane in which the distance between the tips of wings is 50m is flying horizontally with a speed of 360 *km/hr* over a place where the vertical component of earth magnetic field is $2.0\times10^{-4}\ weber/m^2$. The potential difference between the tips of wings would be:

1) 0.1 V
2) 1.0 V
3) 0.2 V
4) 0.01 V

30. A frame CDEF is placed in a region where a magnetic field $\vec{B}$ is present. A rod of length one meter moves with constant velocity 20m/s and strength of magnetic field is one tesla. The power spent in the process is (Take $R=0.2\Omega$ and all other wires and rod have zero resistance)



1) 1 kW
2) 2 kW
3) 3 kW
4) 4 kW

Answer Key

(1)	3	**(16)**	4
(2)	2	**(17)**	4
(3)	2	**(18)**	3
(4)	2	**(19)**	4
(5)	2	**(20)**	4
(6)	2	**(21)**	2
(7)	4	**(22)**	2
(8)	3	**(23)**	1
(9)	1	**(24)**	2
(10)	3	**(25)**	3
(11)	3	**(26)**	2
(12)	2	**(27)**	3
(13)	2	**(28)**	2
(14)	4	**(29)**	2
(15)	2	**(30)**	2